Speaker file 3 – Replay and results

Nov 2, 2026 · @Prakash

One-minute script

You present the last slide and close the talk. About 60 seconds.

> To see what the rover was thinking, we built a **replay tool**. It compiles the real firmware on the laptop and feeds a recorded drive log through it. So for every moment we see where the box believed it was and why it turned.
>
> On the **left** are all the positions the box found with the LiDAR in log_021 and log_022. They stay on the blue path from the start, over the ramp, to M20.
>
> On the **right**: the match score and the heading error over time. On the ramp the LiDAR sees mostly floor, so the box carries on with the gyro.
>
> The **table** compares log_015, before the new controller, with log_020 and 021. Switches between driving and turning halved, from 1.74 to 0.85 per second. Time spent turning dropped from 30 to 9 percent. The jerks from the start-up kick went from 55 percent to about 1. And the rover was located 94 to 97 percent of the time.
>
> **Still open:** side ultrasonics, ramp mode only on the slope itself, and more clearance near mapped walls. Thank you.

What you need to understand

The slide shows that the new controller and the straightened path made driving calmer, and that the box knew where it was almost all the time.

The replay tool (`tools/replay_fw`)

- It compiles the **real** route code from the box (`RouteFollow` , `LocalLocalizer` , `LocalPath.h`) for the laptop, with stub files for the Arduino parts.
- It reads a drive log (`.csv` from the `.bin`) and feeds every sensor sample in, in order. Then it prints what the box would decide.
- Run it with `sh tools/replay_fw/replay.sh cas_box_log/log_NNN.csv` , and plot it with `plot_replay.py` .

- Why it matters: we can find which code block caused a bad turn without driving again. We can also test a change on old logs before flashing.

The v3 logs also record what the box itself believed: position in the leg frame, heading error, match score, and flags for tap, settle, kick, step and located.

Left plot: positions on the map

Each dot is a pose the box found with the LiDAR, moved from its leg frame into the map frame:

```
X = X_k + x\cos\theta_k - y\sin\theta_k, \qquad Y = Y_k + x\sin\theta_k + y\cos\theta_k
```

- Orange: log_021. Purple: log_022 (continues from leg 8 to 9).
- They lie on the straightened blue path from M1, over the ramp at M10–M13, to M20.

Right plot: score and heading error (log_021, legs 1–8)

- **Match score:** how well the scan fits the leg's wall grid (accepted above 0.35).
- **Heading error:** the angle e to the carrot. Most of the time it stays within the ±8° band, with short spikes that are corrected by one tap each.
- On the ramp leg (leg 5, ramp C), 68 % of the scans had fewer than 12 usable LiDAR points. The box kept going on gyro + distance there.

The table, metric by metric

Metric	What it measures	015 → 021	Why it changed
switches/s	changes between driving and turning per second	1.74 → 0.85	one sized tap + wait replaced constant corrections
turning	share of time turning on the spot	30 % → 9 %	8° deadband + straight path
at kick	share of forward driving at the 0.5 kick level (the jerks)	55 % → 1 %	kick only after a real standstill, not after every tap
last leg	furthest leg reached	4 → 8 (10 in log_020)	fewer stops and lost poses

Metric	What it measures	015 → 021	Why it changed
located	share of time with a valid LiDAR pose	– → 97 %	not logged in 015; 94 % in 020

What each log had:

- **log_015:** before the tap controller and the MotorShaper changes;
- **log_020:** new controller, old driven (zig-zag) path;
- **log_021:** plus the straightened path;
- **log_022:** continues leg 8 → 9.

Still open

- **Side ultrasonics unused:** at the start of leg 4, a low board was seen only by the right ultrasonic, and no stop rule uses it.
- **Ramp mode on whole legs:** the LiDAR stop is off for the whole ramp leg, not only on the slope.
- **Walls to 20 cm:** in path mode, walls that are in the map may come to 20 cm (log_020, leg 9, in front of building D).
- **log_023** lost control because weak LiDAR matches turned the heading 30° away from the gyro. This is fixed since with the gyro heading prior.

Likely questions

Did the rover finish the whole route? Not yet. The furthest so far is leg 10 of 17 (log_020). log_021 reached leg 8, and log_022 continued from leg 8 to 9.

How do you know the replay matches the real box? It runs the same source files as the firmware, fed with the same logged sensor data. The v3 logs also contain the box's own pose and heading error, so we can compare both directly.

Why is "located" not 100 %? Some scans see too few walls, for example in open areas or on the ramp, where the LiDAR mostly sees floor. Then the box drives on gyro + distance until a match works again.

What went wrong in log_023? Weak LiDAR matches turned the heading 30° away from the gyro, and the rover steered the wrong way. Now a heading change against the gyro costs match score (heading prior).

Why does fewer switches matter? Every drive/turn switch is a jerk, wastes time and shakes the LiDAR. Halving them made the drive smoother and faster.

What is next? A stop rule for the side ultrasonics, ramp mode only on the slope itself, more clearance near mapped walls, and the rest of the route after leg 10.